ChatGPT

Technical implementation plan (with your decisions)

1) Build outputs

Primary deliverable

- `deck_es_5k_multilingual.csv` (single import into Anki)

QA + transparency

- `qa_report.html` (coverage + error buckets)
- `missing_fields.tsv` (lemma + what's missing)
- `sense_conflicts.tsv` (where cues diverge)
- `build_info.json` (source versions, counts, timestamps)

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2) Data sources (curated only)

2.1 Spanish lemma frequency (base list)

- **Mark Davies 5 000-word list / Anki 5K deck** – We will take the top 5 000 Spanish lemmas from the frequency list compiled by Mark Davies and distributed via the Anki "Top 5000 Spanish Vocabulary" deck. This source already provides lemmas in dictionary form (no conjugation collapsing required) along with their ranking and part-of-speech. Because the deck doesn't supply absolute frequency counts, Zipf scores will either be omitted or approximated from the rank.

Outputs:

- `01_lemmas.tsv` (lemma → pos + rank [+ optional Zipf])

2.2 IPA (curated)

- **Wiktionary** IPA for Spanish entries (dump extraction)

Outputs:

- `02_ipa.tsv` (lemma → IPA)
- `02_wikt_meta.tsv` (pageid/revision/date for provenance)

2.3 Human-curated example sentences

Use one curated example corpus (or two, if licensing allows) **and keep the best match**.

Recommended options (choose based on what you can legally download/use):

- **Tatoeba** Spanish sentences (community-curated)
- **Wiktionary usage examples** (often short and clean)
- (Optional) **OpenSubtitles** is curated-ish but noisy; since you said “only human curated,” treat it as optional only if you personally deem it acceptable.

Outputs:

- `03_examples.tsv` (lemma → example sentence + source)

2.4 Dictionaries for FR/DE (no MT)

- **Primary:** downloadable bilingual dictionary datasets (e.g., TEI/XML such as FreeDict es–fr, es–de)
- **Secondary/fallback:** Wiktionary translations (human edited)

Outputs:

- `04_es_fr.tsv`
- `04_es_de.tsv`
- `04_wikt_fr_de.tsv` (fallback candidates)

2.5 Tamil via dictionary pivot (curated only)

Pipeline:

- Direct: **es→ta** from Wiktionary translations (when present)
- Pivot: **es→en** (dictionary/Wiktionary) + **en→ta** (Tamil lexicon/dictionary dataset)

Key requirement: every edge must be dictionary-derived.

Outputs:

- `05_es_ta.tsv` (direct)
- `05_es_en.tsv`
- `05_en_ta.tsv`
- `05_es_ta_pivoted.tsv`

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3) Internal data model (record per lemma)

LemmaRecord

- `id` (1..5000)
- `es` (lemma)
- `pos`, `rank`, `zipf`
- `ipa`

- `example_es` (+ `example_source`)
- `fr_primary`, `de_primary`, `ta_primary`
- `fr_alt[]`, `de_alt[]`, `ta_alt[]`
- `sources` (compact provenance string)
- `flags[]` (missing_fr, missing_example, sense_conflict, low_confidence_ta, etc.)
- `notes` (formatted alt translations + comments)

Storage: **SQLite** recommended (debuggable, reproducible joins).

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4) Normalization rules (critical for joins)

Apply the same normalization to all Spanish headwords from all sources:

- Unicode NFC
- lowercase
- keep diacritics (ñ, á, etc.)
- strip surrounding punctuation
- normalize whitespace
- remove trailing markers like “(m)”, “(f)”, “v.”, etc. from dictionary entries

Maintain both:

- `es_norm` (for joins)
- `es_display` (original lemma)

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5) Candidate collection & scoring

5.1 Candidate pools per language

For each lemma, gather candidates:

- FR: from es–fr dictionary + Wiktionary translations
- DE: from es–de dictionary + Wiktionary translations
- TA: direct es→ta, else pivot via es→en + en→ta

Keep candidates with metadata:

- `text`, `source`, `sense/gloss if available`, `confidence_base`

5.2 Scoring (simple, deterministic)

Score each candidate using transparent rules:

- +2 if appears in **primary dictionary dataset**
- +1 if appears in **Wiktionary**

- +1 if candidate appears in multiple sources
- −1 if marked archaic/regional (when metadata exists)
- −1 if very long (e.g., > 25 chars) unless it's a standard phrase

- For Tamil pivot:

- direct es→ta starts higher

- pivoted candidates start lower and get flag `ta_pivot`

Pick:

- **Primary** translation = highest score
- **Alt translations** = next best up to N (suggest N=3)

Put alts into `Notes` in a consistent format:

- `FR alt: … | …`
- `DE alt: … | …`
- `TA alt: … | …`

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6) Example selection (human curated)

6.1 Matching

For each lemma:

- First try exact lemma match in sentence (good for nouns/adjectives)

- For verbs, allow inflected forms using a **curated morphology list** if available; otherwise:

- prefer Wiktionary's own usage examples for verbs

- or accept a sentence containing a common inflection if you have a reliable lemma→forms list (must be sourced, not generated)

6.2 Filters

- length ≤ 15 words (configurable)
- avoid proper names when possible
- avoid duplicates across lemmas (keep variety)

If no good example:

- leave blank + flag `missing_example`

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7) Polysemy / sense conflicts handling

Default: one note per lemma

Use the example sentence to anchor sense.

Flag `sense_conflict` when:

- the top FR/DE/TA candidates point to clearly different senses (heuristic: different gloss buckets if provided, or disjoint synonym sets)

Resolution order:

1. prefer candidates whose gloss aligns with example (if gloss exists)
2. prefer candidates supported by multiple sources
3. otherwise keep best primary + put competing candidate into Notes

Only later (optional) split into multiple sense-notes.

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8) CSV export for AnkiMobile

8.1 Final CSV columns

1. `ID`
2. `Spanish`
3. `IPA`
4. `Example`
5. `Tamil`
6. `German`
7. `French`
8. `POS`
9. `FrequencyRank`
10. `Zipf`
11. `Sources`
12. `Flags`
13. `Notes`

8.2 Card templates (HTML-only)

Front:

- `TA:` line
- `DE:` line
- `FR:` line

Back:

- Spanish big
- IPA
- Example
- Notes smaller

No JS. No remote assets.

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9) QA gates (must pass before full build)

Gate A — coverage report

Target minimums (realistic with curated-only constraint):

- IPA coverage: ≥ 85–95%
- FR/DE coverage: ≥ 90%+
- TA coverage: direct+ pivot ≥ 70–85% (depends on en→ta dataset quality)
- Example coverage: ≥ 70–90% (depends on corpus)

Gate B — spot check sample

Random 200 lemmas:

- 1. translation sense matches example reasonably
- 1. Tamil script renders correctly
- 1. no obvious garbage tokens

Gate C — AnkiMobile import test (100-card pilot)

Import first 100 rows:

- field mapping correct
- formatting correct
- review experience sane

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10) Folder structure (concrete)

```
anki-es-5k-multilingual/
  config.yml
  data_raw/
    spanish_5k_deck/
    wiktionary_dump/
    dictionaries/
```

```
    examples/
  data_intermediate/
    01_lemmas.tsv
    02_ipa.tsv
    03_examples.tsv
    04_es_fr.tsv
    04_es_de.tsv
    05_es_ta.tsv
    05_es_ta_pivoted.tsv
  build/
    deck_es_5k_multilingual.csv
    qa_report.html
    build_info.json
  src/
    ingest_5k_deck.py
    parse_wiktionary.py
    parse_dictionaries.py
    select_examples.py
    merge_score_select.py
    export_csv.py
    qa.py
  deck.sqlite
```

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11) Execution sequence (deterministic)

1. `ingest_5k_deck` → `01_lemmas.tsv` (extract and normalize the 5 000-lemma list from the Anki deck)
2. `parse_wiktionary` → IPA + translations tables
3. `parse_dictionaries` → es–fr + es–de + en–ta datasets
4. `select_examples` → `03_examples.tsv`
5. `merge_score_select` → final LemmaRecords (SQLite)
6. `qa` → reports + missing lists
7. `export_csv` → Anki import CSV

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12) Implementation milestones

- **M1:** Lemma list extracted from the Anki 5 K deck, normalized, and frozen (produce `01_lemmas.tsv`)
- **M2:** Wiktionary IPA extraction working
- **M3:** es–fr + es–de dictionaries parsed & merged
- **M4:** Tamil direct + pivot pipeline working with provenance tags
- **M5:** Curated examples integrated + selection heuristics stable
- **M6:** QA gates pass + AnkiMobile pilot import

- **M7:** Full 5k export + documentation
